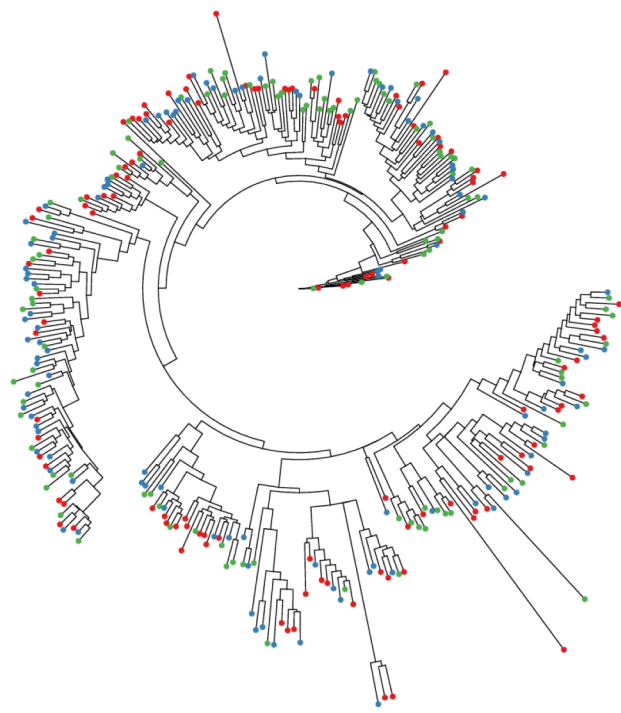


a

Maximum-Likelihood Structural Phylogeny (FoldMason + IQ-TREE)

338 representative sequences from dominant structural class FS_001; tips colored by genome

Q2: FS_001 AA tree (n=338)



genome • PvP01 • PvPAM • PvW1

b

Quantitative Parsimony & Panmixia Analysis

Statistical test of geographic lineage segregation

Continental Tip Distribution:

- PvP01 (Indonesia): 118 tips (34.9%)
- PvW1 (Thailand): 110 tips (32.5%)
- PvPAM (Peru): 110 tips (32.5%)

Total tips: n = 338 (balanced representation)

Phylogenetic Permutation Tests (1,000 Shuffles):

Model Partition	Obs. Score	Null p-value
Partitioned (AA + 3Di)	160	p = 0.55 (NS)
Amino Acid Alone (VT+F+R6)	162	p = 0.69 (NS)
3Di Alone (Garg-Hochberg)	156	p = 0.26 (NS)

- Decisive Biological Conclusions:**
1. Evolutionary Interleaving: Isolates from Asia and South America are closely interspersed throughout the structural tree.
 2. Ancient Pre-Radiation Origin: VIR diversity diversified prior to modern continental spread.
 3. Global Panmictic Repertoire: Subtelomeric ectopic recombination and gene conversion maintain cross-continental antigenic breadth.